

TEST PLAN

VWO Login Dashboard

app.vwo.com

Based on Product Requirements Document – VWO Login Dashboard

1. Objective

The goal of this test plan is to ensure the quality, functionality, security, usability, accessibility, and reliability of the VWO Login Dashboard at app.vwo.com.

The login dashboard is the critical entry point for users accessing VWO's comprehensive suite of experimentation, personalization, and analytics tools. It must deliver a secure, intuitive, and efficient login experience that meets enterprise-grade security standards while minimizing friction for digital marketers, product managers, UX designers, developers, and enterprise teams.

This test plan covers functional correctness, input validation, password management, session handling, multi-factor authentication, SSO, responsive design, accessibility (WCAG 2.1 AA), performance, security compliance (GDPR, OWASP), and integration with the core VWO platform.

2. Scope

Scope of Test Plan for VWO Login Dashboard:

2.1 Functional Testing

- Verify the correctness and functionality of the login form (email + password), Remember Me, Forgot Password, and registration CTA.
- Test primary authentication flow, session management, optional 2FA, and enterprise SSO integration.
- Validate real-time field validation (on blur), email format checks, password strength indicators, and clear error messaging.
- Confirm password reset flow with secure token generation and email-based recovery.
- Verify successful transition to the personalized VWO dashboard after authentication.

2.2 Data Validation Testing

- Ensure the system correctly validates input data and rejects invalid or malformed requests.
- Test boundary values for email and password fields.
- Validate accuracy of success/error responses and feedback messages.

2.3 Error Handling Testing

- Verify appropriate, actionable error messages for failed authentication attempts.
- Check that error responses do not disclose sensitive information.
- Validate graceful handling of unexpected errors and network issues.

2.4 Performance Testing

- Assess login page load time (target: sub-2 seconds on standard connections).
- Measure response time under normal and peak concurrent login loads.
- Validate asset optimization (compressed images, minified CSS/JS) and CDN behavior.

2.5 Security Testing

- Conduct assessments for brute-force protection (rate limiting), session security, and secure token handling.
- Validate HTTPS enforcement, encrypted password storage (industry-standard hashing), and end-to-end encryption of authentication data.
- Check for proper access controls and absence of unauthorized session hijacking vectors.
- Verify compliance with OWASP authentication guidelines.

2.6 Accessibility Testing

- Verify ARIA labels, keyboard navigation, screen-reader compatibility, and clickable form labels.
- Validate High Contrast Mode support and WCAG 2.1 AA compliance.
- Test auto-focus on the first input field and full keyboard accessibility.

2.7 Usability & UX Testing

- Evaluate responsive design across mobile, tablet, and desktop with touch-friendly controls.
- Verify loading states, visual feedback, Light/Dark Mode theme support, and brand consistency.
- Assess clarity of CTAs (free trial signup, Forgot Password) and overall friction reduction.

2.8 Integration Testing

- Verify seamless transition to the VWO Core Platform dashboard after successful login.
- Test analytics integration for login success/failure tracking.
- Validate customer support system integration for login assistance.
- Test enterprise SSO (SAML, OAuth) and optional social login (Google, Microsoft) where applicable.

2.9 Compatibility Testing

- Test across major browsers (Chrome, Firefox, Edge, Safari) and operating systems (Windows, macOS, Android, iOS).
- Validate behavior on different screen sizes and network conditions.

2.10 Regression, Load, Edge Case & Concurrency Testing

- Regression testing after bug fixes or updates to ensure existing login functionality remains intact.
- Load and concurrency testing for thousands of simultaneous login attempts and 99.9% uptime targets.
- Edge-case and boundary testing (extreme inputs, network interruptions, expired sessions).
- Rate-limiting testing to prevent abuse.

2.11 Documentation & Compliance Review

- Assess clarity and accuracy of any user-facing help or recovery documentation.
- Verify GDPR, CCPA, and enterprise audit/security policy adherence.

The scope may evolve based on feedback, changing requirements, or discoveries during testing. It will be reviewed and adjusted to ensure comprehensive coverage of the VWO Login Dashboard.

3. Inclusions

Authentication & Login Process

- Primary email + password authentication with secure validation.
- Session management with configurable timeout.
- Optional Multi-Factor Authentication (2FA).

- Enterprise Single Sign-On (SSO) – SAML, OAuth.
- Remember Me functionality for persistent sessions.
- Auto-focus on first input field.

User Input Validation

- Real-time validation on blur.
- Email format verification (including mobile keyboard support).
- Password strength indicators and complexity enforcement.
- Clear, actionable error messages for failed attempts.

Password Management

- Forgot Password flow with secure token generation.
- Email-based password recovery.
- Enforced password complexity standards.

User Experience & Interface

- Responsive / mobile-optimized design with touch-friendly controls.
- Loading states during authentication.
- Clickable labels and accessibility enhancements.
- Light and Dark Mode theme support.
- Brand-consistent visual design.
- Product announcement banners (where present).
- Clear path to free-trial / account registration.

Security & Compliance

- End-to-end encryption of authentication data.
- Secure password storage (industry-standard hashing).
- HTTPS / SSL-TLS enforcement.
- Rate limiting against brute-force attacks.
- GDPR and enterprise security policy support.
- Session token security.

Performance & Scalability

- Page load within 2 seconds.
- Support for high concurrent login volume.
- CDN and multi-region performance considerations.

User Journeys

- New user discovery → registration CTA → onboarding path.
- Returning user quick login → dashboard transition → context preservation.
- Error recovery and support paths.

4. Test Environments

The following environments, platforms, and configurations will be used:

Environments

Environment	URL / Notes
QA / Staging	Staging equivalent of app.vwo.com (to be confirmed with Dev/DevOps)
Pre-Production	Pre-prod login environment mirroring production configuration
Production	https://app.vwo.com (limited / monitored production verification only)

Browsers & Operating Systems

- Windows 10/11 – Chrome (latest), Firefox (latest), Microsoft Edge (latest)
- macOS – Safari (latest), Chrome (latest)
- Android – Chrome (latest)
- iOS / iPhone – Safari (latest)

Devices & Screen Sizes

- Desktop (1920×1080 and common laptop resolutions)
- Tablet (iPad / Android tablet breakpoints)
- Mobile (common smartphone widths – responsive breakpoints)

Network Conditions

- Wi-Fi / high-bandwidth
- Throttled / 3G-4G mobile simulation
- Intermittent connectivity scenarios for error handling

Access & Roles

- Test accounts with standard user privileges
- Enterprise / SSO-enabled accounts (where available)
- Accounts requiring 2FA
- Tester, Developer, and Stakeholder access as required

5. Defect Reporting Procedure

Defects will be identified based on deviation from the PRD requirements, user-experience issues, security concerns, accessibility failures, or technical errors.

Reporting Steps

- Log defects in the designated defect-tracking tool (JIRA) using the agreed template.
- Provide clear title, detailed reproduction steps, expected vs actual result, environment, browser/OS, and severity/priority.
- Attach screenshots, screen recordings, console logs, or network traces where relevant.
- Link the defect to the related test case / requirement.

Triage & Prioritization

- Severity and Priority assigned during triage.
- Assigned to the appropriate owner (Frontend, Backend, Security, DevOps).

Defect Area	Point of Contact (POC)
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Frontend / UI / UX / Accessibility	To be assigned
Backend / Authentication / Session	To be assigned
Security / Compliance	To be assigned
DevOps / Infrastructure / Performance	To be assigned

Tools: JIRA (primary defect tracking). Communication via daily status updates and end-of-day defect summary emails to stakeholders.

6. Test Strategy

The testing approach follows a structured, risk-based strategy aligned with the PRD phases (Core Authentication → Enhanced UX → Enterprise Features).

Step 1 – Test Design

Test scenarios and test cases will be created for all features in scope. Design techniques include:

- Equivalence Class Partitioning
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing (login states, session timeout, 2FA flows)
- Use Case / User Journey Testing
- Error Guessing & Exploratory Testing

Test cases will be prioritized based on risk, business impact, and frequency of use.

Step 2 – Execution Approach

- Smoke Testing first – verify critical login path is operational. Reject unstable builds.
- Once smoke passes, execute full functional, validation, security, accessibility, and UX suites.
- Multiple testers may execute in parallel across supported environments.
- Defects logged in JIRA; daily status shared with development and stakeholders.
- Types of testing performed: Smoke & Sanity, Functional, UI/UX, Accessibility, Security, Performance (basic), Regression, Retesting, Exploratory, End-to-End user journeys.
- Test cycles repeated until quality gates are met.

Step 3 – Best Practices

- **Context-Driven Testing** – Testing adapted to the login/authentication context and enterprise security needs.
- **Shift-Left Testing** – Early involvement from requirements and design stages.
- **Exploratory Testing** – In addition to scripted cases, exploratory sessions will uncover usability and edge issues.
- **End-to-End Flow Testing** – Full journeys: new user registration path, returning user login, password recovery, SSO, 2FA, and post-login dashboard landing.

7. Test Schedule

Following is the high-level test schedule (exact dates to be confirmed with project timeline):

Task	Target / Notes
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Test Plan Creation & Review	Phase start
Test Case / Scenario Creation	Aligned with Phase 1–3 of PRD
Test Case Execution – Phase 1 (Core Auth)	After stable build for core login
Test Case Execution – Phase 2 (Enhanced UX)	After mobile/accessibility features ready
Test Case Execution – Phase 3 (Enterprise)	After SSO / advanced security available
Regression & Final Verification	Before release candidate
Summary / Closure Reports	End of test cycles

Recommended: 2–3 focused sprints covering the three implementation phases defined in the PRD, plus continuous regression.

8. Test Deliverables

The following artifacts will be delivered:

Deliverable	Description	Target
Test Plan	This document – scope, strategy, schedule, environments, risks	Start of testing
Test Scenarios & Cases	Detailed functional, security, accessibility, UX, and edge-case test cases	Before execution
Defect Reports	Daily/ongoing defects with steps, severity, screenshots	Ongoing
Test Execution Reports	Pass/fail status, coverage metrics	Per cycle
Summary / Closure Report	Bugs by priority/area, residual risk, sign-off recommendation	End of testing

9. Entry and Exit Criteria

9.1 Requirement Analysis / Test Planning

Entry Criteria

- PRD and related design/security documents received and available.

Exit Criteria

- Requirements understood; open questions clarified.
- Test Plan reviewed and approved.

9.2 Test Execution

Entry Criteria

- Test scenarios and test cases signed off.
- Stable build available in the designated test environment.
- Smoke testing passed.
- Test data / accounts (including 2FA and SSO where applicable) prepared.

Exit Criteria

- Planned test cases executed (or justified deferrals documented).
- Critical and high-severity defects resolved or waived with stakeholder approval.
- Test execution and defect reports available.

9.3 Test Closure

Entry Criteria

- Test execution complete; defect reports finalized.

Exit Criteria

- Test Summary / Closure Report delivered and accepted.
- Residual risks documented and accepted by stakeholders.

10. Tools

- JIRA – Defect tracking and management
- Test case management tool (or Excel / structured documents)
- Browser developer tools & network throttling
- Accessibility tools (axe, WAVE, screen readers – NVDA/VoiceOver)
- Performance helpers (Lighthouse, WebPageTest or equivalent)
- Screenshot / recording tools
- Microsoft Word / Excel for documentation and reports
- Mind-mapping tool (optional for scenario design)

11. Risks and Mitigations

Risk	Mitigation
Non-availability of key resources	Backup resource planning and cross-training
Test environment / URL unstable or unavailable	Work on test design, documentation, or other environments; escalate promptly
Insufficient time for thorough testing	Prioritize high-risk areas (security, core auth, accessibility); ramp resources if needed
Late delivery of SSO / 2FA features	Phase testing; mock or isolate dependent tests; re-plan Phase 3
Security / compliance findings late in cycle	Early security-focused testing and regular audits as per PRD risk mitigation
Accessibility gaps discovered late	Include accessibility checks from Phase 2; use automated + manual audits early

12. Approvals

The following documents will be submitted for client / stakeholder approval before proceeding to subsequent phases:

- Test Plan
- Test Scenarios
- Test Cases

- Test Execution & Summary Reports

Testing will advance to the next phase only after the required approvals are obtained.

This Test Plan is derived from the Product Requirements Document for the VWO Login Dashboard and is intended to ensure a secure, performant, accessible, and user-friendly login experience that supports VWO's mission of helping businesses optimize digital experiences.